Roll No.: DL.AI.U4AID24140

Amrita Vishwa Vidyapeetham

Amrita School of Artificial Intelligence, Faridabad

B.Tech Mid-Term Examinations – October 2024

First Semester

Artificial Intelligence & Data Science

23AID101 Computational Thinking

Duration: Two hours | Maximum: 50 Marks

Course Outcomes (COs):

CO	Course Outcomes
CO01	Design and Implement algorithms to solve real-world challenges.
CO02	Develop well-structured and optimized Python programs, utilizing fundamental programming concepts like data structures, control flow, and functions.
CO03	Design modular and reusable code through object-oriented programming to solve complex problems.
CO04	Implement algorithms, and develop solutions for real-world applications, including automation and data analysis, using advanced Python techniques and libraries (e.g., Pandas, NumPy).

Answer all Questions.

Use proper indentation and comments for all Python code snippets.

1 a. Write a Python program to find the sum of all elements in a list using a function. [2] [CO02] [BTL 2]

b. What are the key differences between a list, tuple, set, and dictionary in Python? Explain each data structure in terms of: [8] [CO02] [BTL 2]

- Mutability
- Ordering
- Syntax for creation
- Typical use cases for each

2 a. Write a Python function that takes two lists and returns True if they have at least one common member and print the common member [4][CO02][BTL 2]

b. Write the code to create the dictionary using a dictionary retrieve comprehension from 1 to 5 and the values are their squares. Write a function to retrieve the value for a given key. [6][CO02] [BTL 2]

3 Given the dictionary student_scores = {'Alice': 85, 'Bob': 92, 'Charlie': 78}, write a Python code to:

a. Add a new student David with a score of 88 to the dictionary. [2] [CO02] [BTL 2]

b. Convert the keys of the dictionary into a set. [2] [CO02] [BTL 2]

c. Calculate the average score of all students in the student_scores dictionary. [3][CO02][BTL 2]

d. Identify the student with the highest score and display their name along with the score. [3] [CO02][BTL 2]

4 a. Explain with an example how a recursive function works. Show a simple recursive function and its output. [5] [CO02] [BTL 3]

b. Write a Python program that performs the following operations on a list of integers: [5] [CO02] [BTL 2]

i. Create a list of integers from 10 to 1 in descending order. (1 Mark)
ii. Reverse the list to make it ascending. (1 Mark)
iii. Count how many times the number 5 appears in the list. (1 Mark)
iv. Create a new list containing only the even numbers from the original list and print it. (2 Marks)

5 A bookstore wants to maintain its inventory using Python. The inventory includes information about the book title, author, and quantity in stock. You need to write Python programs to manage the following (Each sub question carries 2 Marks). [10][CO01][BTL 3]

a. Create a dictionary where the keys are book titles, and the values are dictionaries with keys "author" and "quantity". Write a function to add a new book to the inventory.
a. Write a function that updates the quantity of a specific book title when new stock arrives.
b. Write a program to display all books in the inventory where the quantity is below a certain threshold (e.g., less than 5).
c. Write a Python function to delete a book from the inventory if its quantity becomes zero.
d. Write a function that prints all book titles and authors sorted alphabetically by title.

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Course Outcome /Bloom's Taxonomy Level (BTL) Mark Distribution Table

CO	Marks	BTL	Marks
CO01	**10**	BTL 1	-
CO02	**40**	BTL 2	**40**
CO03	-	BTL 3	**10**
CO04	-	BTL 4	-